INCHPOUND

A-A-59483
3 March 2004

COMMERCIAL ITEM DESCRIPTION

OVERALLS, POWERSTRETCH WITH "RAINBOW DROP SEAT"
SPEAR System: Layer-3

1. SCOPE

This Commercial Item Description covers the requirements for the Layer 3 Medium Weight Powerstretch Bib Overall. The powerstretch bib overall is intended to use by male and female military personnel of the Department of Defense as a multi-purpose lightweight insulation layer of the Extended Cold Weather Clothing System (ECWCS).

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-COCT (Bldg 6), 700 Robbins Ave., Philadelphia, PA 19111-5092 or emailed to leo.bailey@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil

AMSC N/A FSC 8415

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2. CLASSIFICATION

The Powerstretch Bib Overall with Rainbow Drop Seat will be of one type, color Black 470 or Camouflage Green 483 in color, as specified and in the following sizes as specified:

SCHEDULE OF SIZES

Small- Regular Medium-Regular Large-Regular Large-Long
Extra Large-Regular Extra Large-Long

3. SALIENT CHARACTERISTICS

3.1 ***Design and construction***. The one-piece, full length bib overall has a two-way front zipper, mid-chest inner pockets, full-length side zippers, rainbow drop seat, reinforced seat and knees. All joining seams shall be flat locked construction to reduce bulk and provide comfort in movement.

3.2 ***Basic material***. The following material shall be used to provide the environmental protection, user comfort; lightweight and durability, and dimensional moisture management needed for wear in field, combat and operations. The basic material for the Bib Overall shall be a circular knit nylon and spandex jersey plaited to a single sided velour, mid- weight Polartec Powerstretch Malden Mills Style 9401M®. This material provides the required military appearance and functionality required by the military. The fabric shall conform to the following requirements:

TABLE I. ***Knit cloth physical requirements.***

CHARACTERISTIC	REQUIREMENT	TEST METHOD
Weight (oz. per square yard) (oz. per linear yard)	7.1 ± 0.71 12.0 ± 1.2	ASTM D3776
Colorfastness: Laundering Crocking Light Carbon Arc Xenon	 2 min Nylon, Acetate, 3 min all other fibers Dry – 4.0 / Wet 3.0 20 Hrs. – Excellent 4-5 85 Kj. – Excellent 4-5	 AATCC 61-Option 2a cold wash, 3 cycles AATCC 8 AATCC 16 Option A and Option E
Dimensional Stability – Avg. Pilling	Wales: 5.0% max. Courses: 7.0% max. 4 minimum	AATCC 135-3 cycles, III, A, iii ASTM D-3512, Random Tumble Method
Stretch	Wales: 65% min. Courses: 65% min.	ASTM 2594, Form fit

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3.2.1 ***Internal pocket material***. The material for the mid chest inner pockets shall be a warp knit tricot mesh of 100% CoolMax® polyester or equal. The fabric shall conform to the following requirements:

TABLE II. ***Pocketing material.***

CHARACTERISTIC	REQUIREMENT	TEST METHOD
Weight (oz. per square yard)	2.0 ± 0.2	ASTM D3776
Colorfastness:		
Laundering	4, minimum	AATCC 61, IIA
Crocking	4, minimum	AATCC 8
Perspiration	4, minimum	AATCC 15
Light	4, minimum, after 40 hrs.	AATCC 16E
Dimensional Stability-Avg.	5.0% Wales X 5.0% Courses max.	AATCC 135, III, A, iii

3.2.2 ***Reinforcement, rainbow drop seat and knees material***. The material for the rainbow drop seat and knee patches shall be 2-way stretch Spandura Style 3400®. The fabric shall conform to the following requirements:

TABLE III. ***Reinforcement material.***

H.Warshow & Sons, Inc. Fabric Style 3400-44®
Description: WARP KNIT TRICOT

CONSTRUCTION

Fiber Content: 91% Cordura® Nylon / 9% Lycra® Spandex
Fabric Weight (oz. / yd.²): 9.4 5%
Warp – Wales / Inch 36 ± 2 Fill – Courses / Inch 29 ± 2

TEST PROPERTIES				TEST METHOD
Colorfastness to Laundering (Gray Scale Readings-GSR)				
Shade Change:	3-4			AATCC 61 "2A Wash"
Staining:	3-4			AATCC 61 "2A Wash"
Colorfastness to Crocking				
Dry:	3-4			AATCC 8
Wet:	3-4			AATCC 8
Colorfastness to Perspiration (GSR):	3-4			AATCC 15
Shrinkage:	Warp (Wale) Direction:	8% max.		AATCC 135
	Fill (Course) Direction:	8% max.		120º Wash / Tumble Dry
Stretch:	Zwick Tester – CRE	20#	Effective Load	
Warp (Wale) Direction:	135%	± 10%	3 Cycles	
Fill (Courses) Direction:	75%	± 10pts.	3 Cycles	

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3.2.3 ***Thread***. Thread for needle and bobbin (looper) shall be commercial 100% textured polyester thread, approximate Tex size 35, 2 or 3 ply conforming to A-A-52095 with minimum breaking strength 2.5 lbs. The thread shall be non-staining and show good colorfastness to laundering. The thread color shall match the color of basic material.

3.2.4 ***Labels***. Each pair of overalls shall have a commercial size label, an identification label, and a care label. The size and identification labels may be combined. The identification label shall contain the item description, contract number, NSN number, fiber content information, and the contractor's name. The size label color shall be white. The inscription shall have a minimum font size of 10 points. The inscription legibility, label and label attachment shall last the expected life of the garment. If available, a basic material supplier care label and hang tag may be used, except the information must be as cited in paragraph 3.2.4.1.

3.2.4.1 ***Care Label***. The care/instruction label for Powerstretch Bib Overall shall include the following information:

COLD WASH
LINE DRY.
DO NOT USE BLEACH.
AVOID USE OF
FABRIC SOFTENERS.
DO NOT IRON.
DO NOT DRY CLEAN.

3.2.4.2 ***Size Label***. The size label shall contain, as a minimum, the following information, as applicable:

SMALL REGULAR

OVERALLS, POWERSTRETCH, LAYER 3
64% Polyester, 24% Nylon, 12% Spandex Knit
CONTRACT #: xxxxxx-xx-x-xxxx
NSN #: xxxx-xx-xxx-xxxx
Contractor's Name

3.2.4.3 ***Label/Tag***. Each pair of overalls shall have a bar-code label affixed to the garment. The barcode label shall be clearly legible and readable by scanner as well as human readable. The bar coding element shall be a 13 digit national stock number (NSN). The label shall comply with the DOD LOGMARS symbology. It must be located so that it is completely visible when the garment is folded and /or packaged and shall cause no damage to the item.

3.2.4.4 ***Label Placement***. The labels shall be securely stitched to left inner pocket (as worn) so as not to damage the inner mesh pocket of the garment, or diminish the operational effectiveness of the pocket.

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3.2.5 ***Slide Fastener*:** All slide fasteners used in the overalls shall conform to A-A-55634. All sliders used shall have an opening large enough to accommodate a 3/8 inch wide thong.

3.2.5.1.1 ***Side Opening Slide Fasteners***. The slide fastener used in the full length side closure disengages at the ankle and shall be a continuous element polyester monofilament, 2-way separator with double sliders back to back, auto lock, size 5-7 coil, 9/16 inch wide tape.

3.2.5.1.2 ***Center Front Slide Fastener***. The center front closure shall be a continuous element polyester monofilament closed bottom with double sliders, auto lock, size 5-7 coil, 9/16 inch wide tape.

3.2.5.1.3 ***Rainbow Drop Seat Slide Fastener***. The rainbow drop seat slide fastener shall be chain continuous element polyester monofilament, size 5-7 coil, 9/16 inch wide tape, with double sliders.

3.2.5.2 ***Tape for Slide Fastener Thong and Hanger Loop***. The slide fastener thongs shall be flat nylon tape, with ends knotted and a minimum of 3 inches in finished length. The hanger loop shall finish 2-3/4 inches ± 1/4 inch in length. The color shall match the basic material.

3.2.6 ***Elastic Webbing***. The elastic webbing for the mesh pocket opening shall be 3/4 inch wide.

Size	Inches
X-Small	7 ¾
Small	8 ½
Medium	9 ¼
Large	10
X-Large	11

3.2.7 ***Binding.*** The binding for around the neck and armholes shall be a tricot knit weighing 4.1 +0.2 oz. per sq. yd. with a fiber blend of 90% nylon and 10% Lycra, slit into a 1-1/2 inch wide binding strip. The color shall match the basic material.

3.2.8 ***Fastener Tape, Hook and Loop***. The hook and loop fastener tapes for the inner chest pockets opening shall be 1/2 to 3/4 inch wide, conform to type II, class 1 of A-A-55126. The hook and loop tape shall be 2 inches ± 1/4 inch in length.

3.3 ***Patterns.*** The government shall furnish a complete set of patterns or a master pattern with grade rules, to maintain uniformity and consistency in manufacturing. The government patterns shall be used to create the contractor's working patterns. Minor modifications are permitted to accommodate manufacturing procedures, however the design and finished measurements shall be maintained.

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TABLE IV. ***List of pattern parts***

Pattern Abbreviation	Nomenclature	Cut Parts
Basic Fabric, Power Stretch®:		
FRONT	Front	2
UPPER BACK	Upper Back	2
DROP SEAT	Drop Seat	2
BACK LEGS	Back Legs	2
ZIPPER FLAP	Zipper Flap	1
Warp Knit Tricot, Spandura:		
DROP SEAT PATCHES	Drop Seat Patches	2
KNEE PATCHES	Knee Patches	2
Mesh, 100% COOLMAX:		
INNER CHEST POCKETS	Inner Chest Pockets	2

3.4 ***Configuration.*** The following specifics are needed to provide uniform appearance, comfort and durability in use during field duties, combat, and operations other than war.

3.4.1 ***Seaming.*** The seams shall be consistent, exhibit a uniform appearance and conform to ASTM D-6193, Stitch and Seam Types listed in the Constructions of Bib Overall Table V below. All material edges shall not ravel.

TABLE V. ***Construction of bib overall***

No.	Operation	Stitch Type	Seam Type
01	Stitch knee patch on each front panel as indicated by marks on pattern.	301	LSd-1
02	Over edge elastic to top edge of mesh pockets.	504	SSa-1
03	Fold elastic to inside of pocket and top stitch.	301	Efb-1
04	Stitch long hook tape on inside of pocket opening.	301	SSs-2
05	Stitch size and care labels on the left pocket.	301	Lsbj-1
06	Stitch pockets to body in alignment with loop tape.	301	Lsd-1/SSs-2
07	Stitch edge of reinforcement patch to drop seat panels.	301	SSa-1
08	Join drop seat panels.	607	FSa-1
09	Join upper back panels.	607	FSa-1
10	Set slide fastener to drop seat curve.	602	LSa-2
11	Stitch 2 inches of binding material at each end of drop seat slide fastener.	301	SSq-2
12	Attach legs to bottom of drop seat.	607	FSa-1
13	Join shoulder and front crotch seams	607	FSa-1

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14	Join inseam.	607	FSa-1
15	Over edge raw edges of overall sides, front opening and slide fastener extension (flap).	504	EFd-1
16	Set front slide fastener and extension. Turn and topstitch.	301	LSbu-2a/b
17	Set side slide fasteners 1/2 inch below edge of the underarm. Turn and topstitch.	301	LSbu-2a/b
18	Hem bottom of legs as indicated by notches.	605	EFa-2
19	Bind the neck catching the hanger loop at center back. Bind the armholes.	406	BSc-2
20	Finish ends of binding tape by folding under 1/2 inch ±1/8 inch to the inside of the overalls and bartack.	Bartack	Bartack
21	Attach zipper thong tapes and knot ends	-	-

4. REGULARITY REQUIREMENTS

4.1 ***Recycled, Recovered, or Environmentally Preferable Materials***. Recycled, recovered, or Environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. PRODUCT CONFORMANCE

5.1 ***Product conformance***. The products provided shall meet the salient characteristic of this commercial item description, and shall conform to the cited patterns, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance.

5.2 ***Component and end item inspection***. In accordance with 3.2 and 5.1, components and end item shall be tested in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable procurement documents. The Government reserves the right to inspect all components and end item to determine conformance to requirements. Lot inspections shall be in accordance with ANSI Z1.4.

5.4 ***End item visual examination***. The Powerstretch Bib Overall shall be examined for the major defects listed below.

TABLE VI. ***End item examination***

Examination	Defect
Materials	Any hole, cut, tears, run, needle chew, drill hole, thin place, dye streak, misweave affecting appearance or serviceability or weakening defect in the material, any mend or patch, shade variation within a part or between parts, color not as specified.
Seams and stitching	Seam: puckered, distorted, pleated, wavy,

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	twisted, irregular or open, loose stitch tension resulting in loose seam, visible mend, edge or raised stitching sewn too close to edge, resulting in damage to cloth; seam type not as specified, raw edges; any part of Overalls caught in any unrelated stitching, the edge of any component part required to be forced out having folds of more than 1/8 inch. Stitches: broken or missing thread or stitch, stitch type not as specified.
Workmanship	Any component part omitted or not as specified, distorted, full, tight or twisted parts. Any defect not specified in this table which affects, form, fit or function.
Cleanness	Spot, stain, excessive thread ends not trimmed or removed, raw edges, odor, affecting appearance and serviceability.
Labels	Omitted, incorrect, illegible, not attached specified; bar codes omitted, not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code not visible on folded, packaged item; bar code causes damage to the Overall.
Dimensions and finished measurements	Not as specified.
Packaging	Any Overall not packaged in accordance with the contract or purchase

5.5 ***Finished dimensions***. The Bib Overalls shall conform to the dimensions listed in the Table VIII below:

TABLE VII. ***Powerstretch bib overalls finished measurement***

SIZE	**1/2 - CHEST (A)**	**1/2 - SEAT (B)**	**1/2 -BOTTOM-LEG (C)**	**INSEAM (D)**
SMALL	18”	18”	4-1/2”	28”
MEDIUM	20”	20”	4-3/4"	29”
LARGE	22”	22”	5”	30”
X-LARGE	24”	24”	5-1/4”	31”
LARGE-LONG	22”	22”	5”	33”
X-LARGE-LONG	24”	24”	5-1/4”	34”
TOLERANCE	± 1/2”	± 1/4”	± 1/4”	± 1”

NOTE: All measurements shall be taken with the garment laid out flat under no tension and with all slide fasteners closed.

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(A) Chest – Measure at base of armseye, from folded edge to folded edge, with garment laid flat.
(B) Seat - Measure 7 inches up from crotch seam, from folded edge to folded edge.
(C) Bottom-Leg - Measure bottom of overall leg, from folded edge to folded edge.
(D) Inseam - Measure from crotch seam to bottom edge of hemmed leg.

6. PACKAGING

6.1 ***Packaging, packing and marking***. The packaging and marking used shall be in accordance with the contractor's standard commercial practice. Each bib overalls shall be packed individually in a plastic bag and twenty (20) bib overall shall be packed in each box. If needed, mixed sizes and small boxes may be used to complete a shipment.

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory unless otherwise stated in the contract or purchase order.)

7.1 ***Intended use.*** The overalls are for wear by military personnel of the Department of Defense. They are intended for Special Operations Forces Personal Equipment Advance Requirements (SPEAR). They are used as an insulation layer of the Extended Cold Weather Clothing System in field, combat, and operations other than war.

7.2 ***Acceptance criteria***. Sampling for inspection shall be in accordance with ANZI/ASQC Z1.4. The end items shall be examined for visual defects listed in paragraph 5.4. The lot size shall be expressed in units of overalls. The sample unit will be one pair of overalls. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects. The end item shall be examined for dimensional conformance in accordance with paragraph 5.5. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of overalls. The sample size shall be on pair of overalls. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. Requirements for this acceptance criteria shall be referenced in the contract or purchase order.

7.3 ***Government documents***.

7.3.1 ***Specifications, standards, and handbooks***. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Acquisition Streamling and Standardization Information System (ASSIST) database and supplement thereto, cited in the solicitation.

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SPECIFICATIONS

CIDS

A-A-55634	- Zippers (Fasteners, Slide Interlocking)
A-A-55126	- Fastener Tape, Hook and Loop, Synthetic
A-A-52095	- Thread, Polyester, Textured

Source of Government documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

7.3.2 ***Source of non-Government documents***. The following documents form a part of this document to the extent specified herein.

ASTM International

ASTM D2594	Standard Test Method for Stretch Properties of Knitted Fabrics Having Low Power
ASTM D3512	Standard test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Random Tumble Pilling Tester
ASTM D3776	Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM D6193	Standard Practice for Stitches and Seams

(For all inquires please contact the American Society For Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428). Website address http://www.astm.org

AATCC

AATCC 8	Colorfastness to Crocking: AATCC Crockmeter Method
AATCC 15	Colorfastness to Perspiration
AATCC 16	Colorfastness to Light
AATCC 61	Colorfastness to Laundering, Home and Commercial: Accelerated
AATCC 135	Dimensional Changes in Automatic Home Laundering of Woven and Knit Fabrics

(For all inquires please contact the American Association of Textile Chemists and Colorists (AATCC), P.O. Box 12215, Research Triangle Park, NC 27709-2215.) Website address http://www.aatcc.org

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AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQC Z1.4 – SAMPLING PROCEDURES AND TABLES FOR INSPECTION OF ATTRIBUTES

(For all inquires please contact the American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036). Website address http://www.ansi.org.

MILITARY INTERESTS:

Custodians
Army - GL
Navy - MC

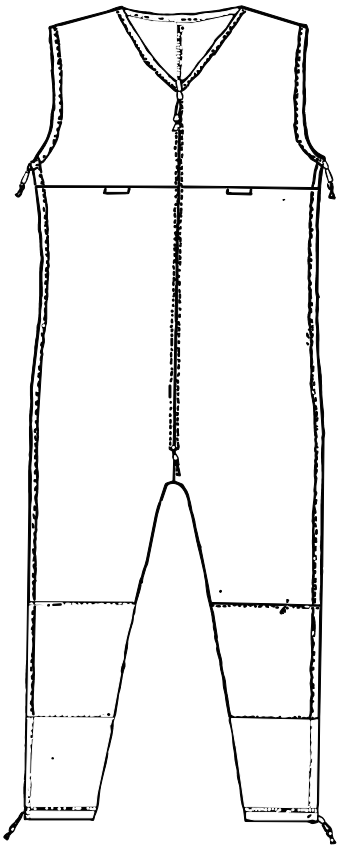
CIVIL AGENCY COORDINATING ACTIVITY:
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PREPARING ACTIVITY:
DLA - CT

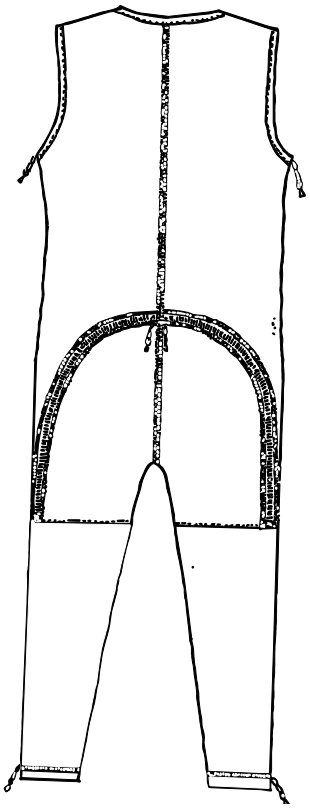
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FRONT VIEW



BACK VIEW

FIGURE I - POWERSTRETCH BIB OVERALL

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